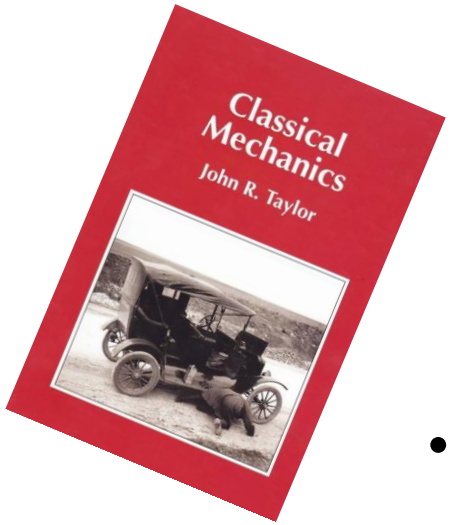
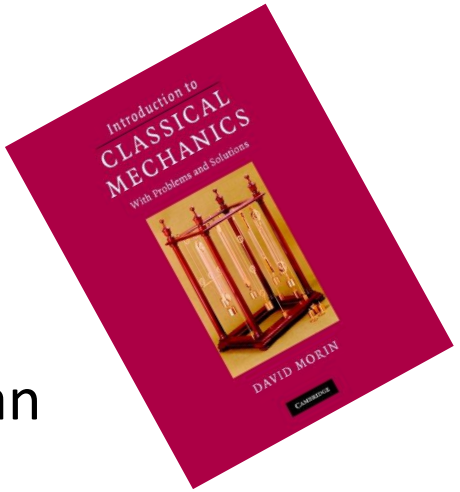


PHYSICS I
WEEK 1

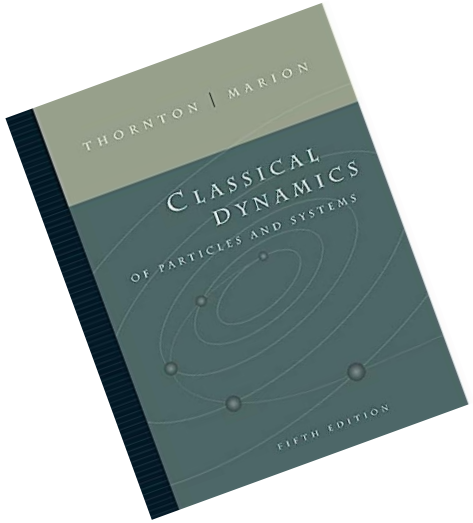
Siddharth Dhomkar
Department of Physics
IIT Madras
sdhomkar@smail.iitm.ac.in
Office: HSB 116B

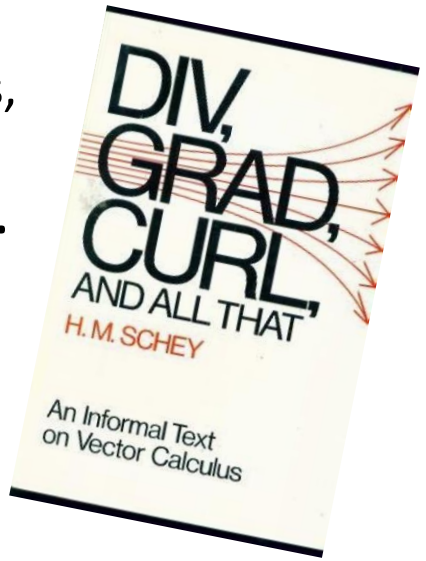


Course Information



- The grading will be based on two quizzes (2×20%), and an end-of-semester exam (60%).
- Reference books:
  - Classical Mechanics, **J. R. Taylor** (University Science Books, 2005)
  - Classical Dynamics of particles and systems, **S. T. Thornton and J. B. Marion** (2008)
  - Introduction to Classical Mechanics with problems and solutions, **D. Morin** (Cambridge UK, 2007)
  - Div, grad, curl and all that: an informal text on vector calculus, **H. M. Schey** (3rd ed., Norton and Co.)







Course Information

- **First day**: 1st August 2024, **Last day**: 12th November 2024
- ~ 18 Classes before Quiz 1 (2nd September)
- ~ 20 Classes between Quiz 1 & Quiz 2 (10th October)
- ~ 16 Classes between Quiz 2 & End Sem (20th November)
- **~ 45 Lectures and 9 tutorials**
- **Slot:** D1 and D2

Course Information

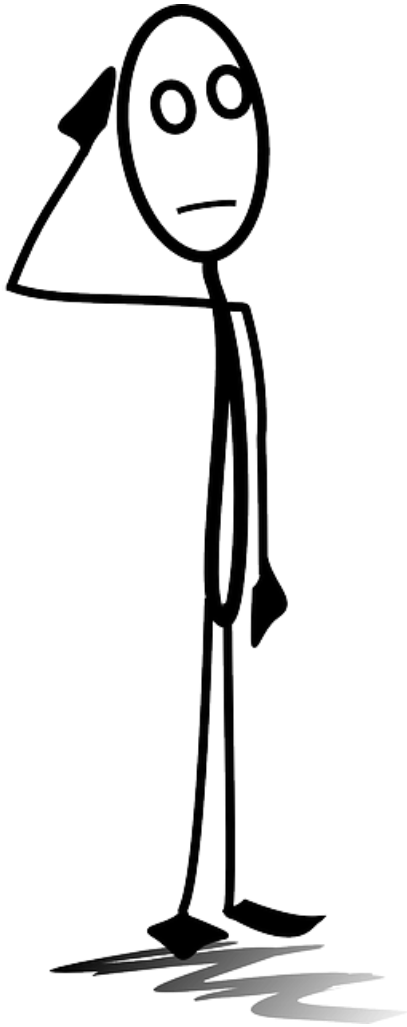
DAY	1 8.00 – 8.50	2 9.00 – 9.50	3 10.00 - 10.50	4 11.00 – 11.50		5 13.00 - 13.50	6 14.00 - 14.50	7 15.00 – 15.50	8 16.00 –16.50	Extra 17.00 – 17.50
Mon	A1	B1 MA1101 RJN 101	C1	D1 PH1010 RJN 101	Lunch recess 11.50 am – 13.00 pm	G2	B2	E2	C2	XX ID1200 CRC 101-103 & CRC 201-205
		P2					P1			
Tue	B1 MA1101 RJN 101	C1	D1 PH1010 RJN 101	E1		A2	C2	D2	E2	YY
		Q2					Q1			
Wed	C1	D1 PH1010 RJN 101	E1	F1		B2 MA1101 RJN 101	D2	E2	F2	XX
		R2					R1			
Thu	E1	F1	G1	A1		D2 PH1010 (Tutorial) RJN 101	F2 MA1101 (Tutorial) RJN 101	G2	A2	YY ID1200 CRC 101-103 & CRC 201-205
		S2 PH1030					S1			
Fri	F1	G1	A1	B1		C2	F2	A2	B2	XX
		T2 GN1101 (FFT Terrace)					T1			

RMN – Raman Block | CRC- Classroom Complex | RJN – Ramanujan Block

WEEK 1: OUTLINE

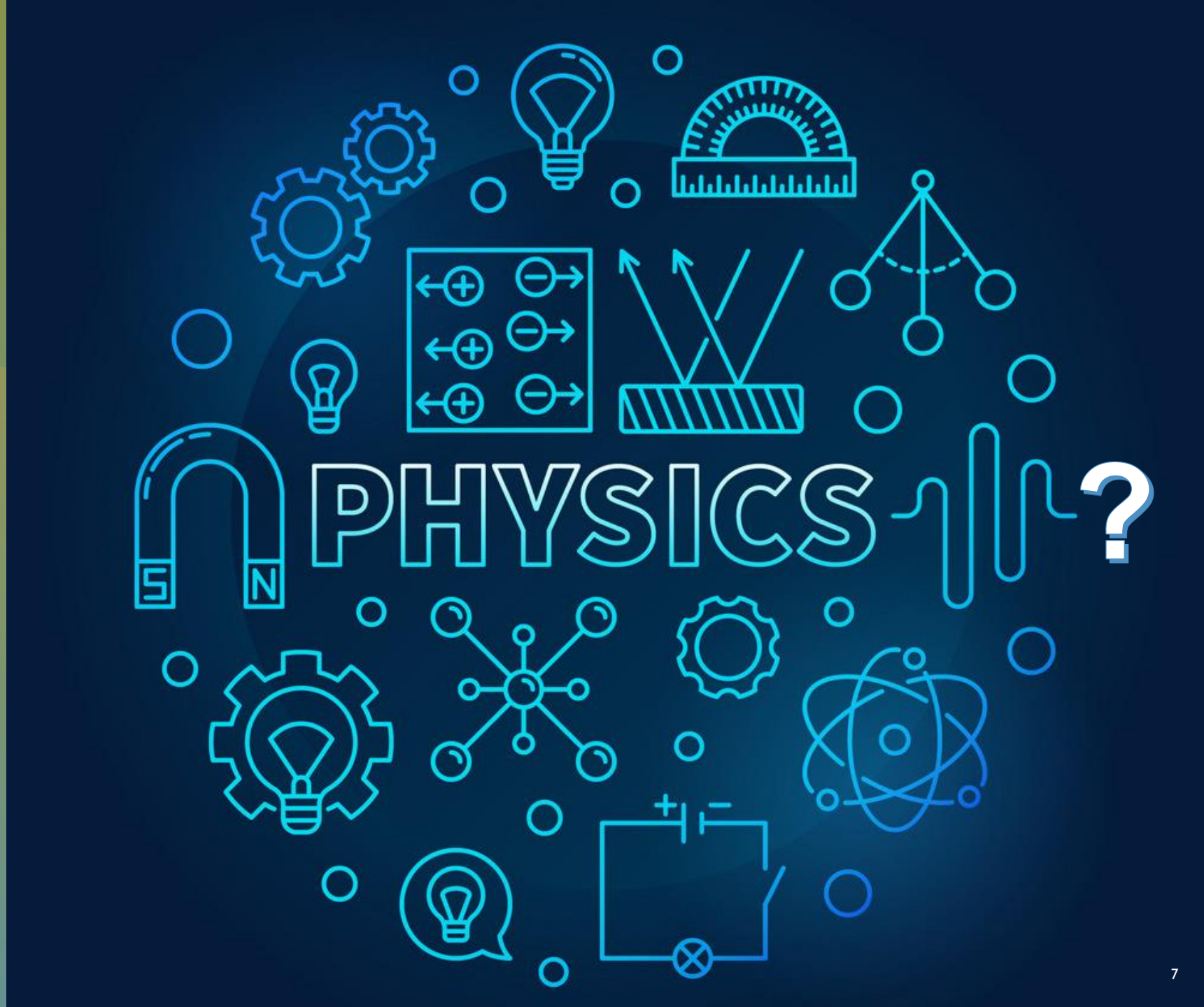
Unit 1 – Newton's Laws:

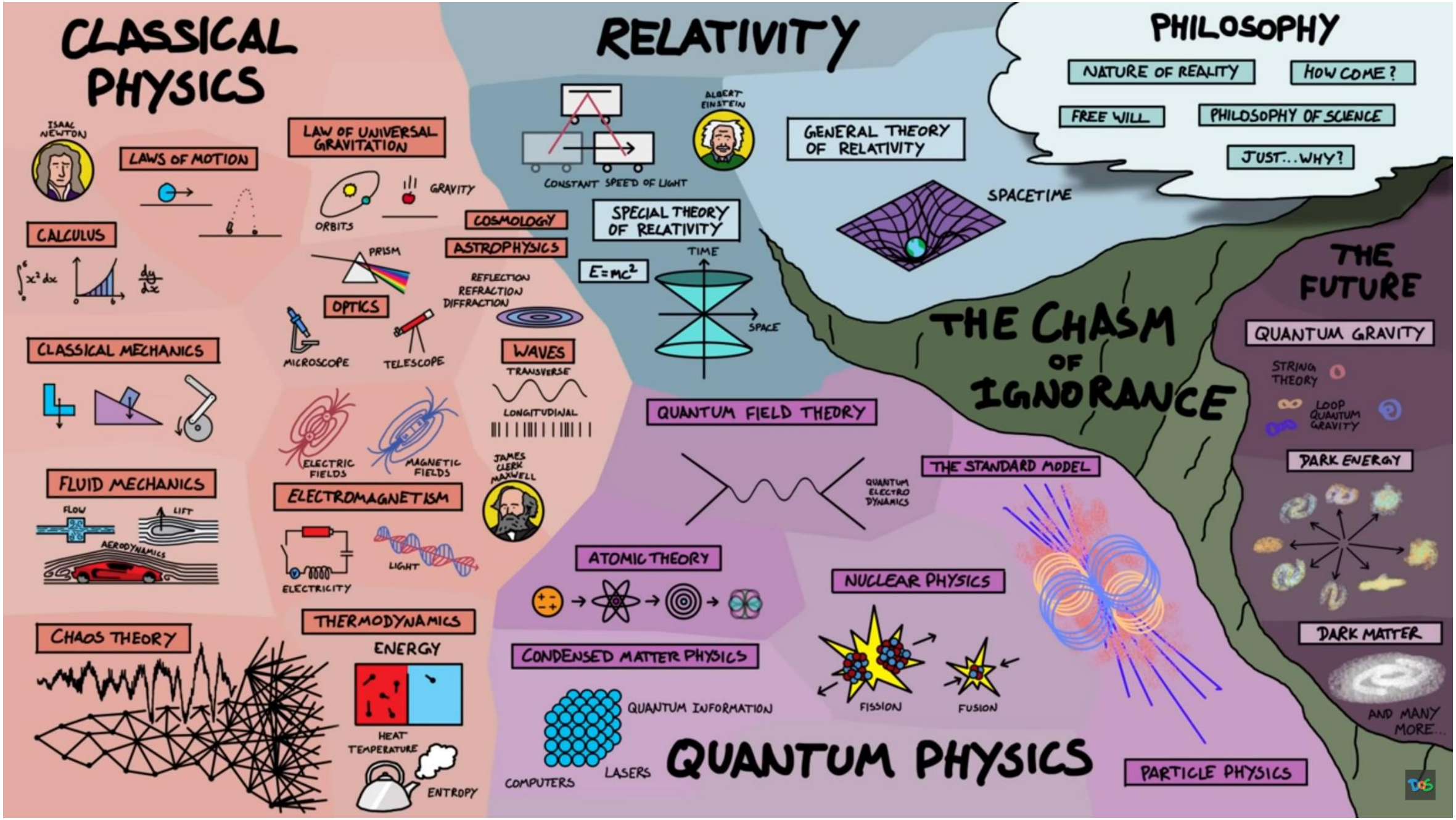
- Fundamental concepts – Space, time, mass, and force : Taylor 1.1, 1.2, 1.3
- Dot-product, Cross-product: Taylor 1.2




LET'S GET STARTED!

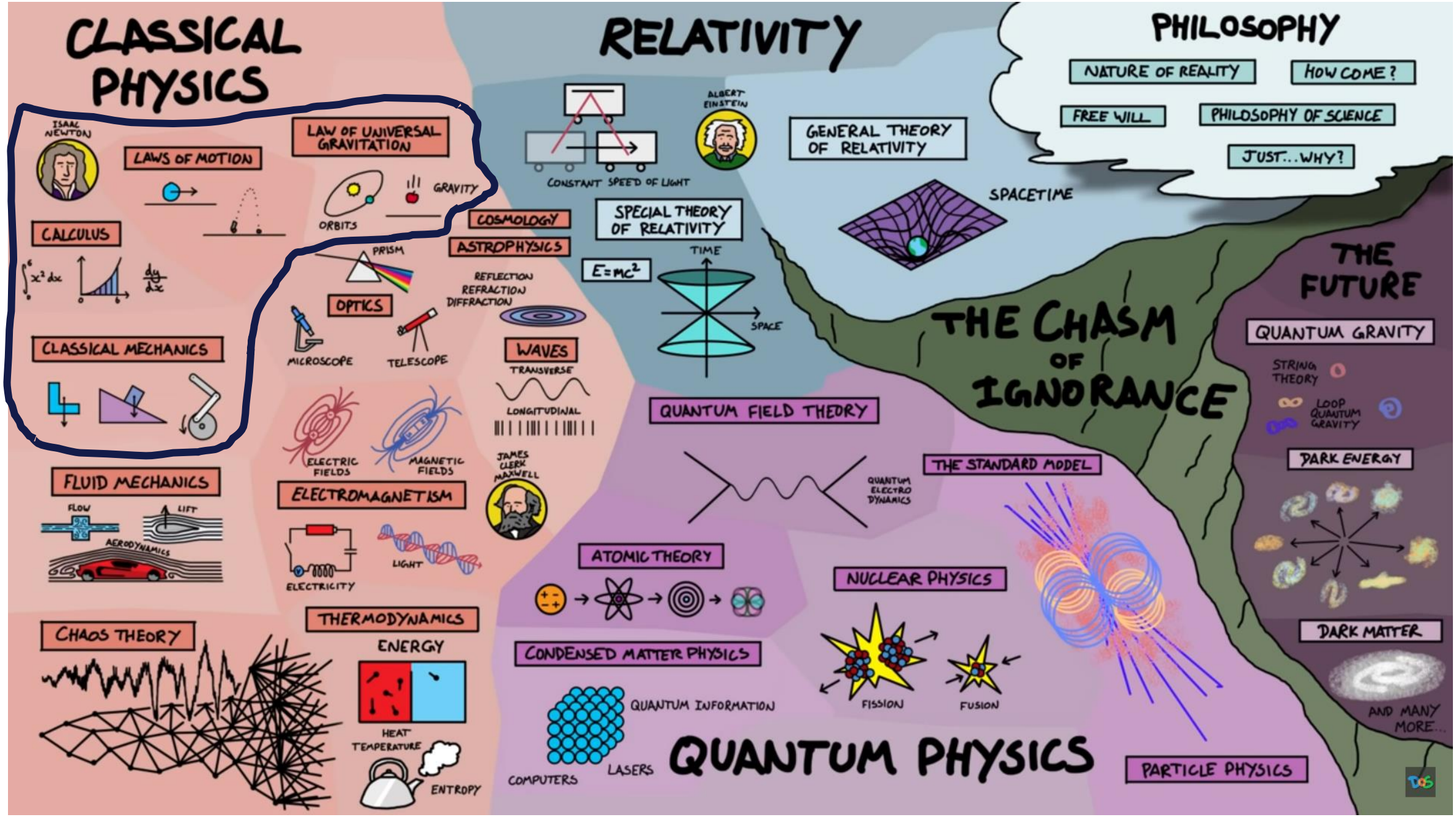
What is





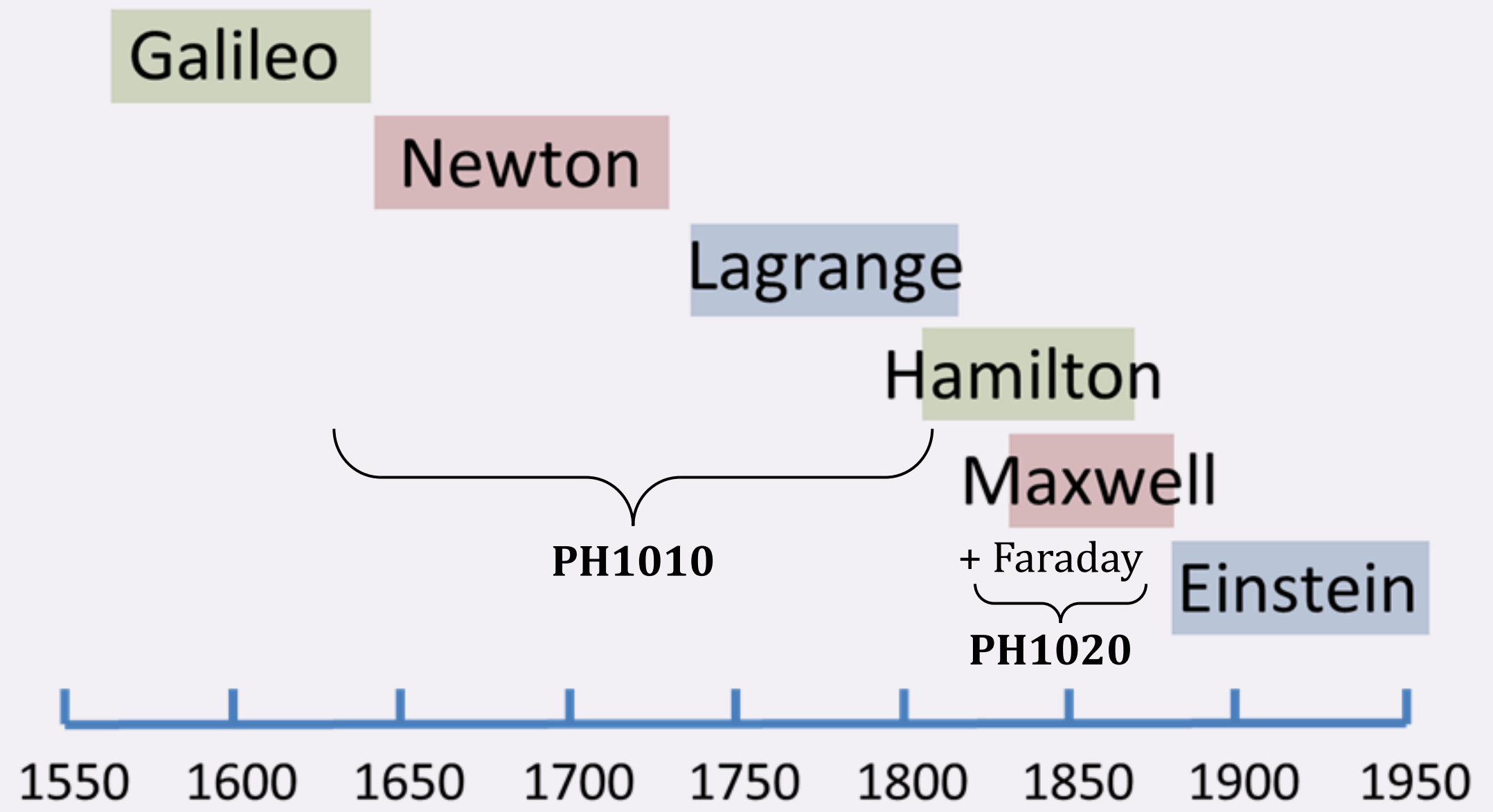
[Domain of Science - The Map of Physics]

Domain of Science
1.26M subscribers

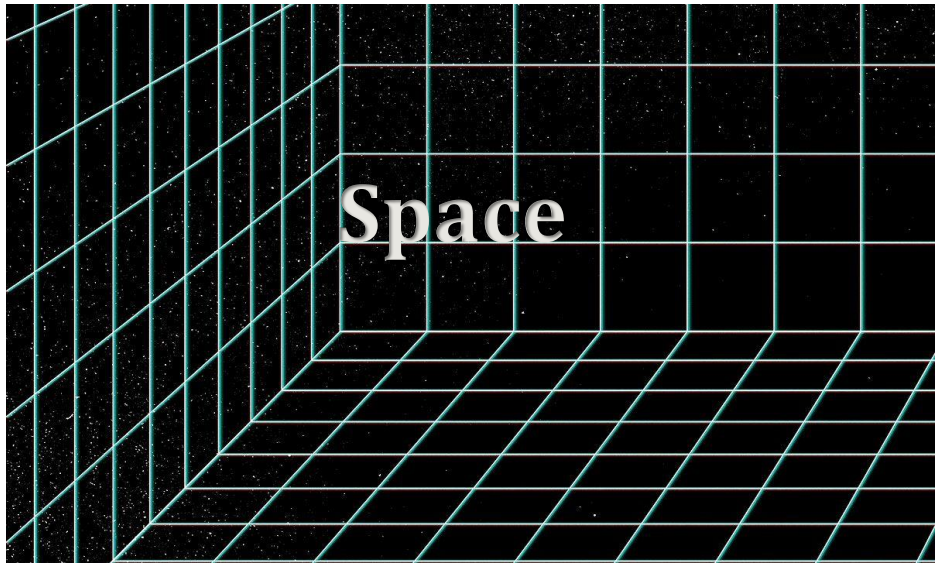


[Domain of Science - The Map of Physics]

Domain of Science
1.26M subscribers

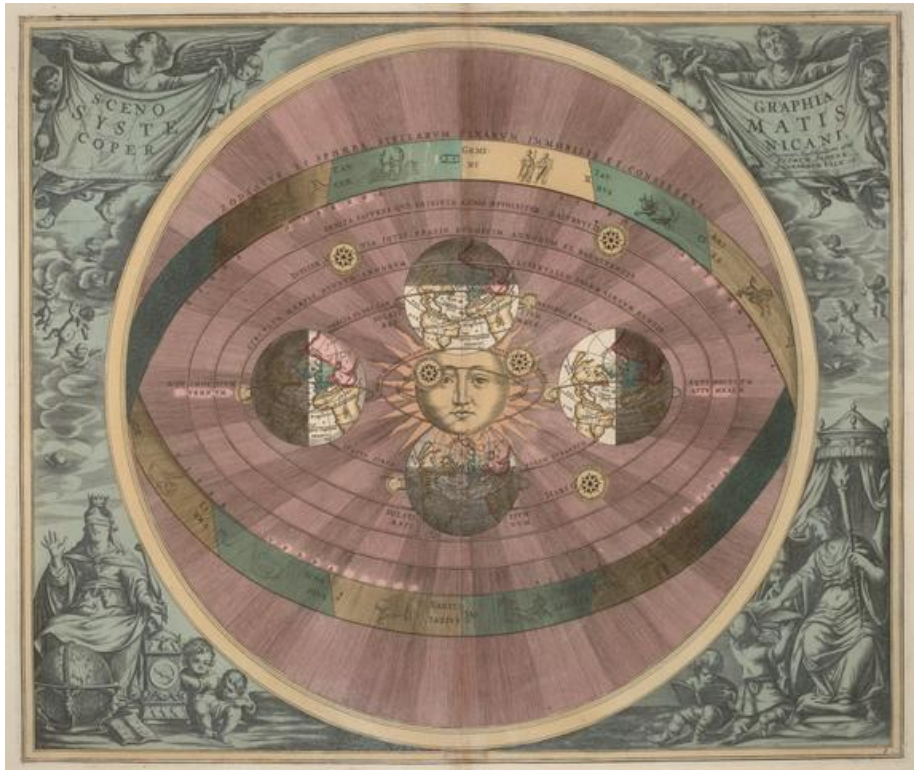

Galileo
Newton
Lagrange
Hamilton
Maxwell
Einstein
PH1010
+ Faraday
PH1020
1550
1600
1650
1700
1750
1800
1850
1900
1950

REVISITING NEWTON'S LAWS



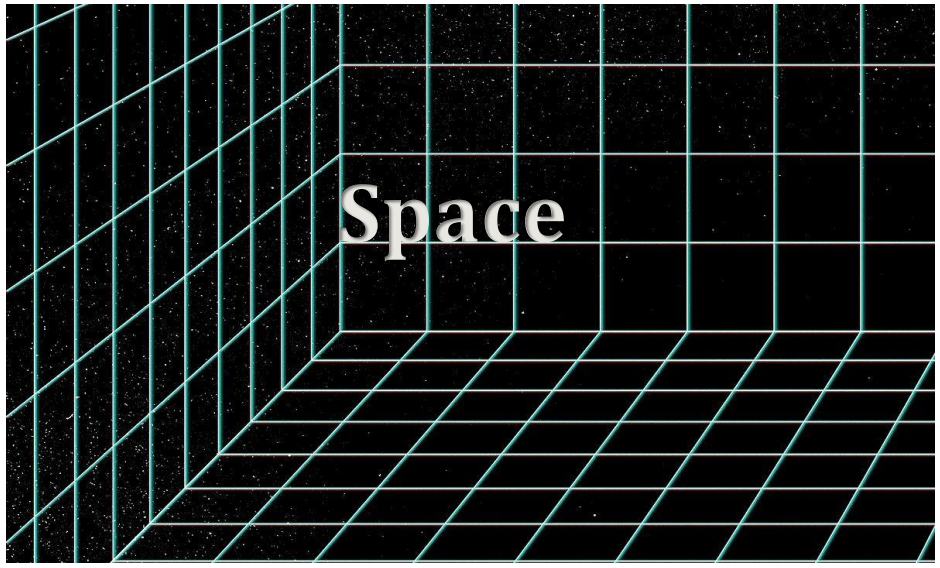


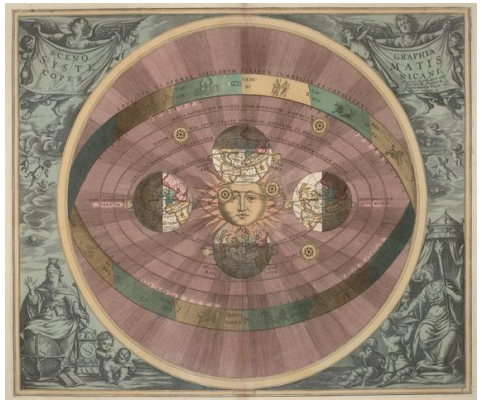
REVISITING NEWTON'S LAWS





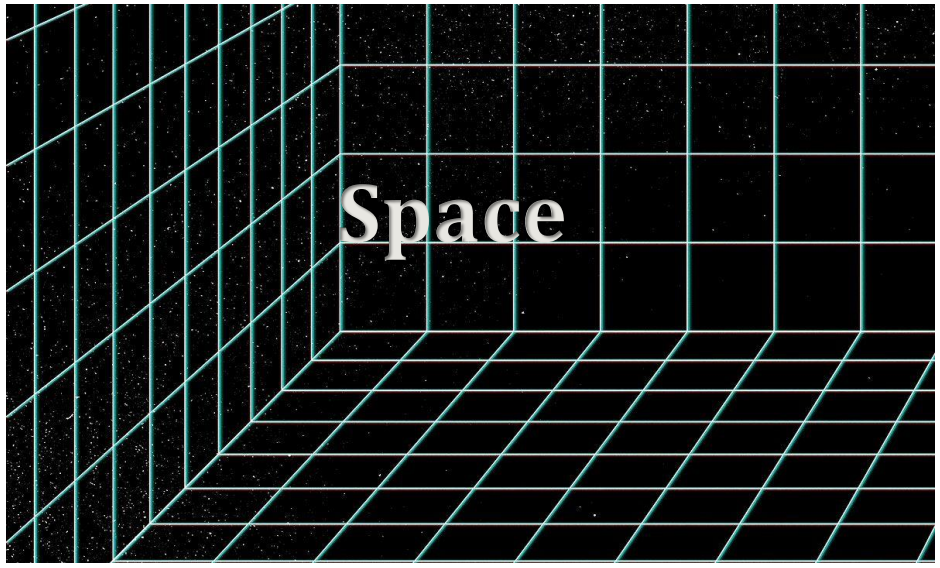
REVISITING NEWTON'S LAWS

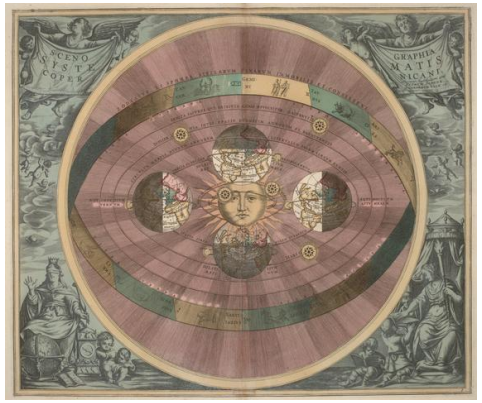






REVISITING NEWTON'S LAWS: VECTORS





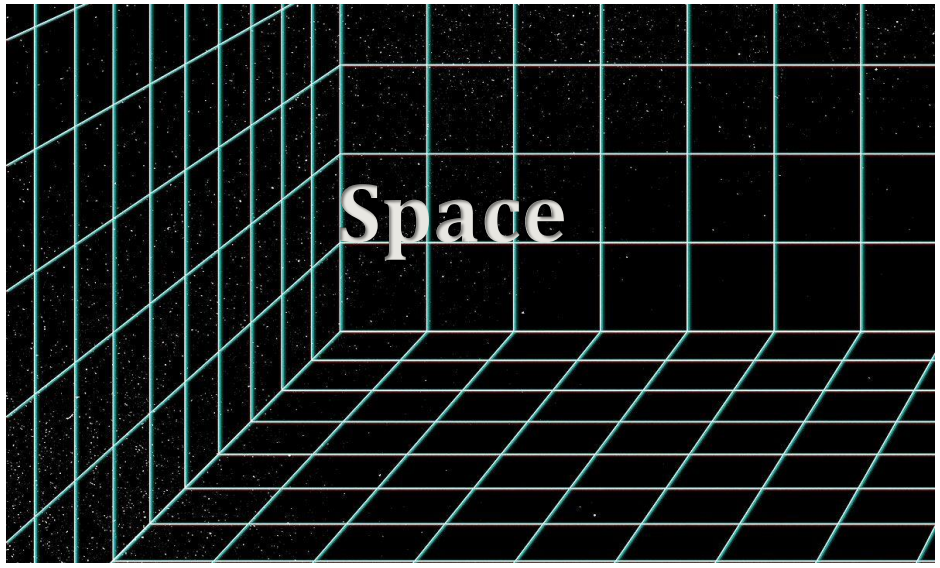


Scalar

$$F = mg$$ ← Vector



REVISITING NEWTON'S LAWS: VECTORS

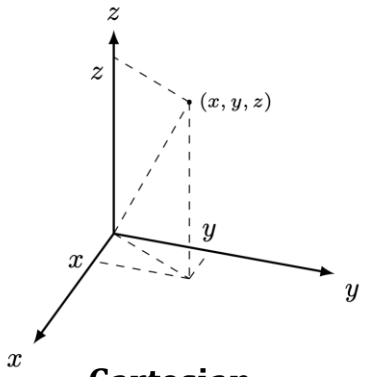




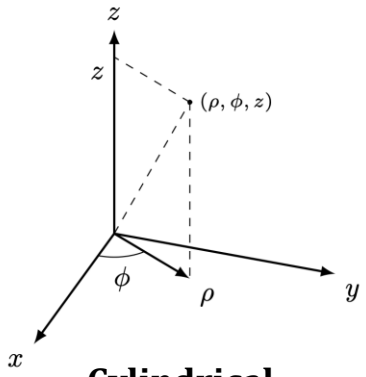
Scalar

$$F = mg$$ ← Vector

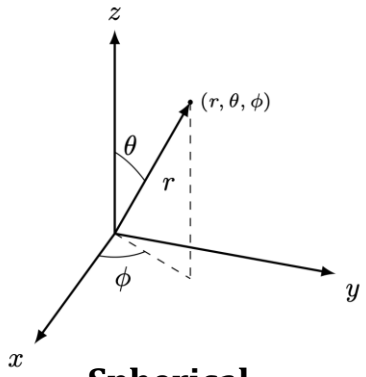




Cartesian



Cylindrical



Spherical

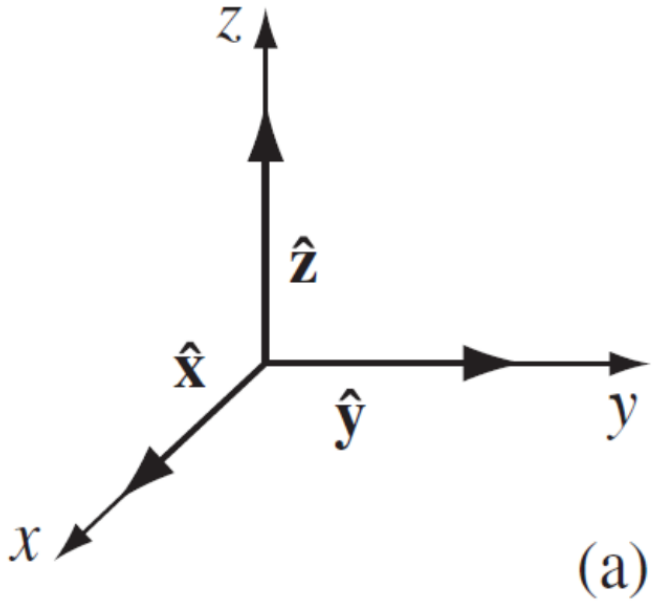
Position Vector

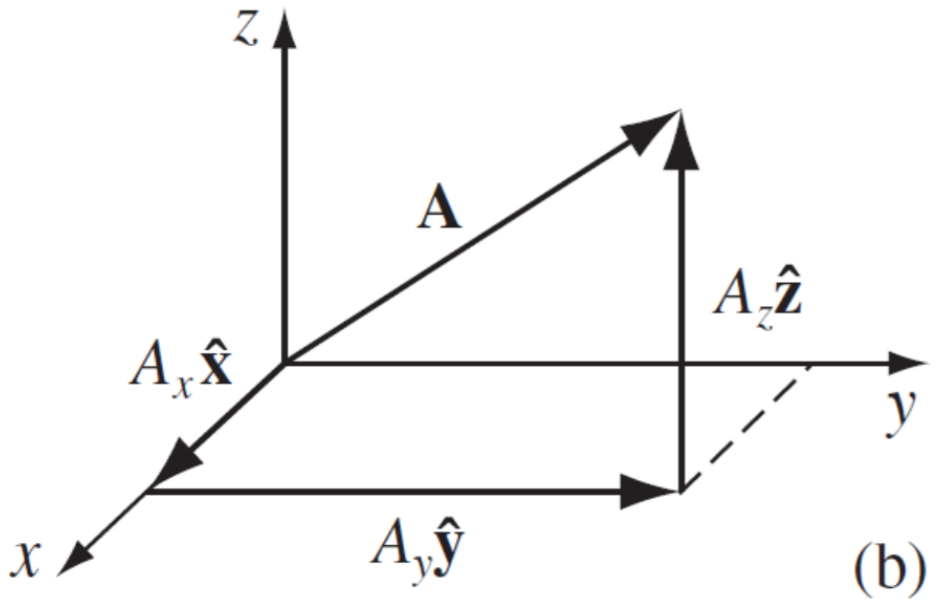
The three numbers (x, y, z) specify the vector fully

$$\mathbf{r} = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}},$$

$\hat{x}$, $\hat{y}$ and $\hat{z}$ are mutually perpendicular unit vectors

REVISITING NEWTON'S LAWS: VECTORS





$$\mathbf{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}$$

$\hat{e}_x = \hat{\mathbf{x}},\ \hat{e}_y = \hat{\boldsymbol{y}},\ \hat{e}_z = \hat{\boldsymbol{z}}$

REVISITING NEWTON'S LAWS: TIME

What is time?

Oxford Dictionary: The indefinite continued progress of existence and events in the past, present, and future regarded as a whole

REVISITING NEWTON'S LAWS: TIME

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REVISITING NEWTON'S LAWS: TIME

What is time?

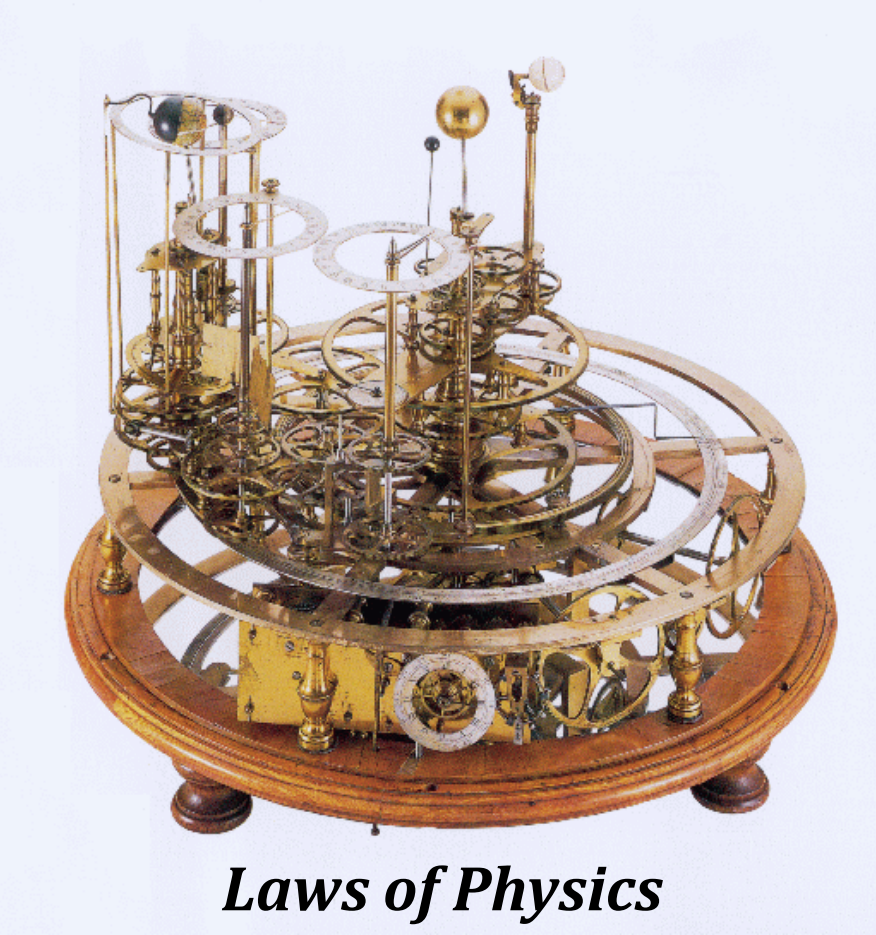
Oxford Dictionary: The indefinite continued progress of existence and events in the past, present, and future regarded as a whole



The classical view is that time is a single universal parameter t on which all observers agree. That is, if all observers are equipped with accurate clocks, all properly synchronized, then they will all agree as to the time at which any given event occurred.

REVISITING NEWTON'S LAWS: TIME

Determinism & Clock-work universe



Laws of Physics

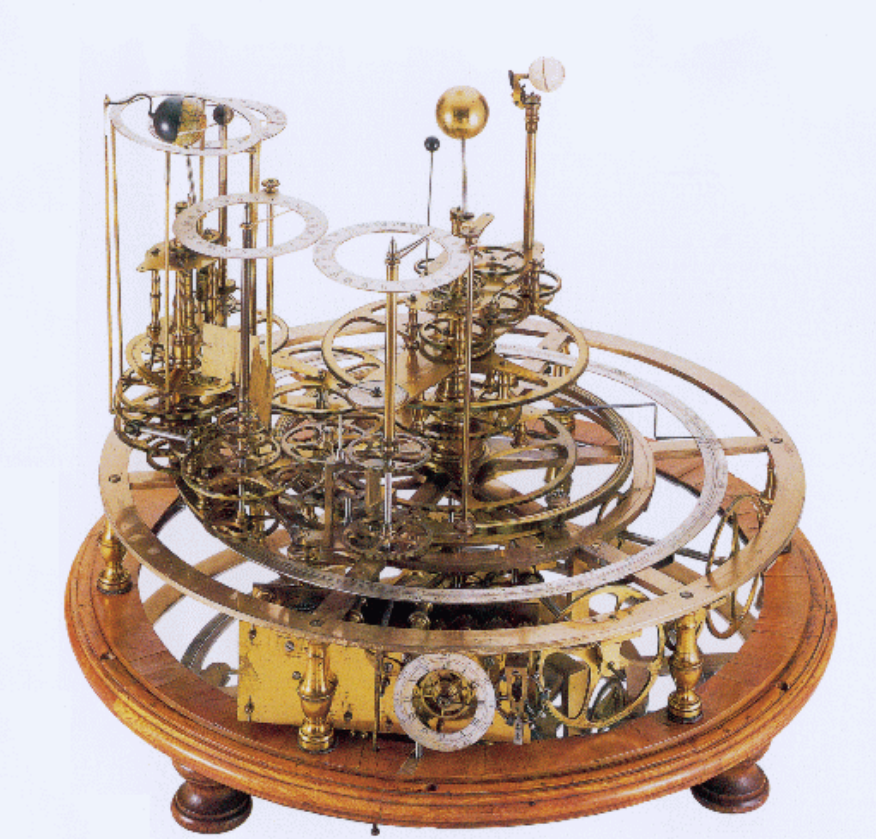
REVISITING NEWTON'S LAWS: TIME

Determinism & Clock-work universe

Input:

Initial Conditions

$x(t_0)$ & $p(t_0)$



Laws of Physics

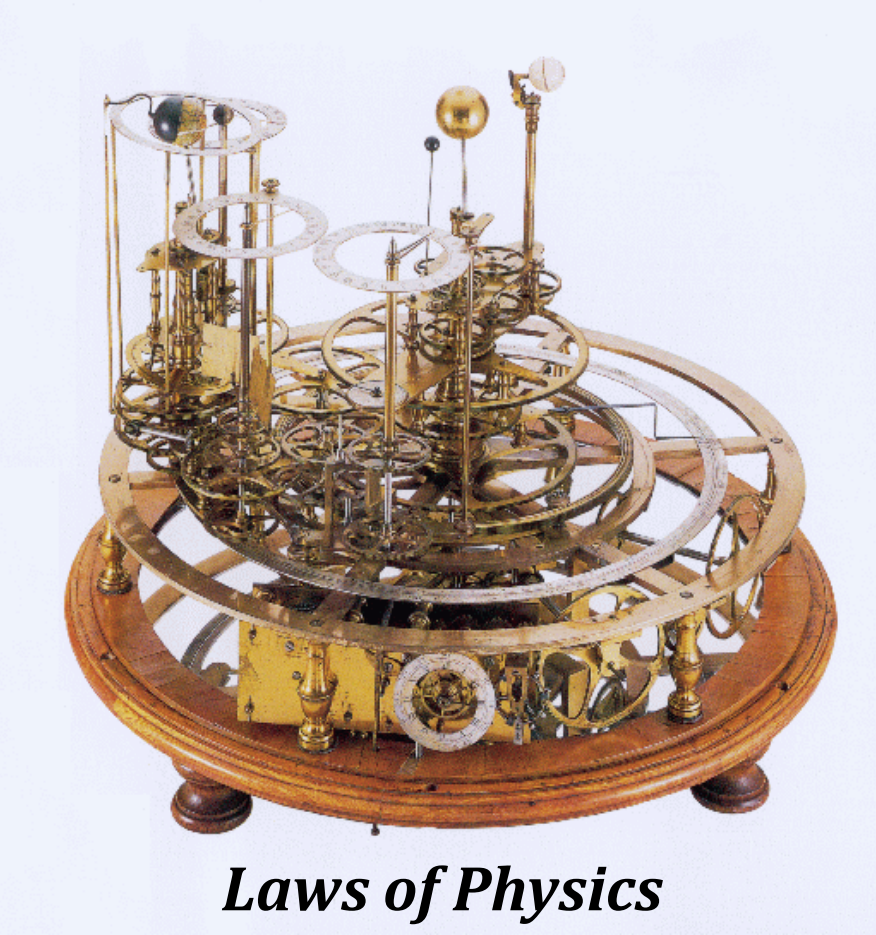
REVISITING NEWTON'S LAWS: TIME

Determinism & Clock-work universe

Input:

Initial Conditions

$x(t_0)$ & $p(t_0)$



Laws of Physics

Output:

Final state

$x(t_f)$ & $p(t_f)$

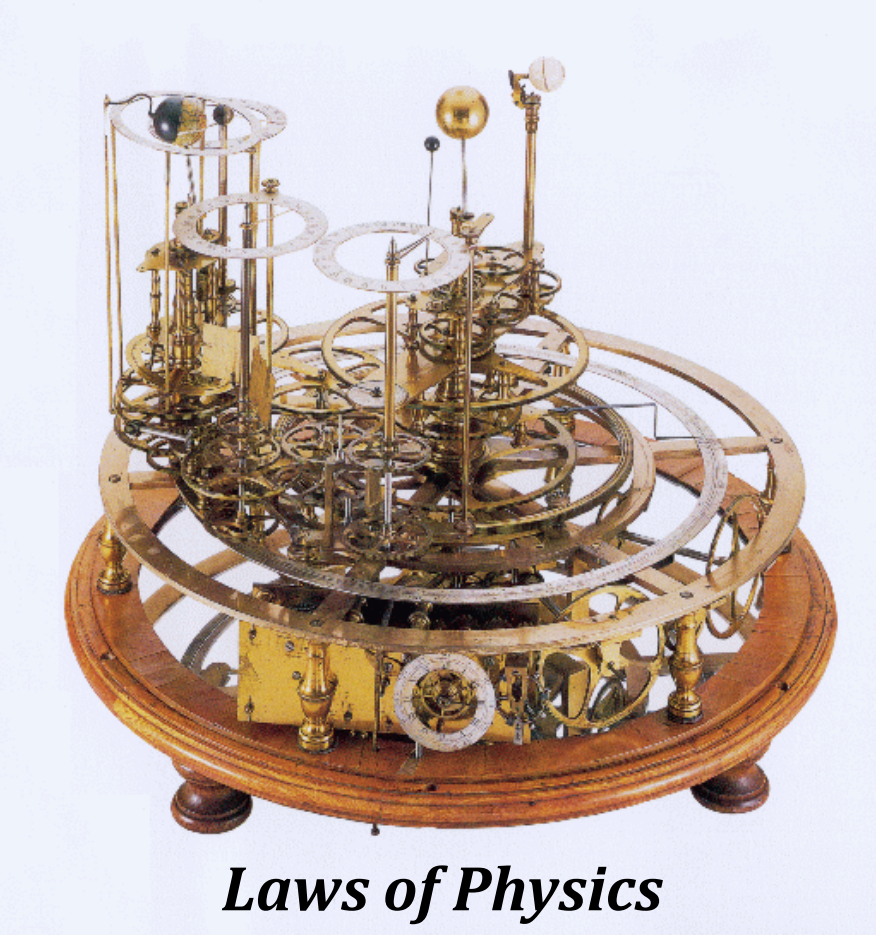
REVISITING NEWTON'S LAWS: TIME

Determinism & Clock-work universe

Input:

Current state

$x(t_f)$ & $p(t_f)$



Laws of Physics

Output:

Initial state

$x(t_0)$ & $p(t_0)$

REVISITING NEWTON'S LAWS: TIME

Determinism & Clock-work universe

Momentum Before

Equals



Fully reversible!